

## DETAILED TEST ANSWER - QUESTION 4

Assessment question: *Analyze scan results showing ports as Open, Closed, and Filtered. Explain what each state indicates.*

**Assessment hint:** Open = service running; Closed = reachable but no service; Filtered = firewall blocks probes.

### Introduction

Port scanning is used to determine the network state of ports on a target host. After a scan, ports may be reported as Open, Closed, or Filtered. Understanding these states helps a penetration tester identify available services, confirm host reachability, and recognize when a firewall or other security control is affecting the scan.

### 1. Open Port

**Meaning:** An open port means the target host is reachable and an application or service is listening on that port. The service is accepting the type of connection or probe being tested.

- **Security meaning:** An open port represents an exposed service and becomes part of the system's attack surface.
- **Next step:** Identify the service and, when authorized, determine its version and configuration so possible vulnerabilities can be assessed.
- **Example:** If TCP port 22 is open, the host may be running SSH. The tester should then review whether SSH is properly secured and up to date.

### 2. Closed Port

**Meaning:** A closed port means the host is reachable and responding, but no application is currently listening on that port. Therefore, the port is not providing an active service at the time of the scan.

- **Security meaning:** A closed port normally exposes less direct service risk than an open port because no application is accepting connections there.
- **Important point:** Closed does not mean the host is unreachable. The response actually provides evidence that the host can be reached.
- **Example:** A TCP port can respond in a way that confirms the host is reachable while showing that no service is listening on that port.

### 3. Filtered Port

**Meaning:** A filtered port means that the scanner cannot determine clearly whether the port is open or closed because a firewall or another filtering mechanism is blocking, dropping, or interfering with the probe.

- **Security meaning:** The result often indicates an access-control rule, firewall, ACL, or similar security device is controlling the traffic.
- **Important point:** Filtered does not prove that no service exists. It only means that the scanner cannot reliably determine the state from the response it receives.
- **Example:** A firewall may silently drop probes to a protected service. Because the scanner receives insufficient information, it reports the port as filtered.

## 4. Direct Comparison

**Open:** The port has a service listening and accepting connections. This is the main state that reveals an exposed service and normally deserves further assessment.

**Closed:** The host is reachable, but no service is listening on that port. It confirms reachability but does not show an active service.

**Filtered:** A firewall or other filtering mechanism is preventing the scanner from determining whether the port is open or closed.

## 5. How the Tester Should Analyze the Results

The three states should not be treated as equal findings. An open port usually deserves the most attention because it exposes a live service. A closed port mainly confirms that the host is reachable without exposing an active service. A filtered port shows that a security control is affecting the scan, so the tester must not assume that the service is absent.

For filtered ports, an authorized tester can compare results from appropriate scan methods and review firewall or ACL behavior. The purpose is to understand the network controls and avoid making an incorrect conclusion from an incomplete scan result.

## 6. Analysis Example

Suppose a scan reports port 22 as Open, port 25 as Closed, and port 443 as Filtered:

- **Port 22 - Open:** A service is listening. The tester should identify the service and assess its configuration and security.
- **Port 25 - Closed:** The host is reachable, but no service is listening on that port at the time of the scan.
- **Port 443 - Filtered:** A firewall or other security control is preventing the scanner from deciding whether the HTTPS service is available.

## 7. Importance in Penetration Testing

Correctly interpreting port states is important because it prevents false conclusions. Open ports help reveal the real services exposed on a host. Closed ports help confirm host reachability. Filtered ports provide information about defensive controls and indicate that the scan result may be incomplete. Together, these results help the tester build an accurate picture of the target network.

## Conclusion

In conclusion, an **Open** port means a service is running and accepting connections, a **Closed** port means the host is reachable but no service is listening, and a **Filtered** port means that a firewall or other filtering mechanism prevents the scanner from determining the exact state. Therefore, the tester should prioritize open services for further assessment, recognize closed ports as evidence of reachability, and investigate filtered results without assuming that no service exists.

**Exam keywords:** Open = service running | Closed = host reachable, no service | Filtered = firewall/filtering blocks or interferes with probes